3508 -Rear Gear Gp. Inspection & reuse criteria

(Re-use & salvage guideline SEBF9244) **DOC # IMS/ GCPL/ CRC/ ENG/34**

Customer :	**Job no.**		
Engine sr.	**Model :**		
SMU :	**Acceptable**	**Reject**	
A. Cluster shaft (1011378) O.D : 74.99 ± 0.010 mm.			
No seizer mark & hot spots should be visible			
No scoring mark should be there			
No fretting mark on the back side plate			
Oil holes MUST be clean & clear			
No CRACK at shaft & plate weld joint			
Shaft area should not be cleaned with emery paper			
Replace **4 bolts (8M2530) as MANDATORY**			
(Torque 105± 7 lb-ft) **(SENR 5189)**			
B. Cluster Gear assy. – 1011377			
Check both the gear (35 & 92 teeth) profiles for scoring , damage, broken teeth etc.			
Used bush (**4P5460**) should be cleaned with soft cloth only. No nick, dent, crack , temper color, scoring mark should be visible. Bush ID MUST be within **75.06 ±0.010** mm at Front & Rear end.	**AA**	**BB**	**CC**
	A1A1	**B1B1**	**C1C1**
If bush ID exceeds limit. Option 1 : Replace entire assy (1011377) with new			
Option 2 : Install new semi finish bush (4P5460) & finish diam. (refer CAT guidelines for m/cing job) . Re-check ID as per spec.			
Replace **4 bolts (0L1351) as MANDATORY**			
Thrust Plate 1011365 is free from scoring , damage etc.			
******** CLEARENCE BETWEEN CLUSTER BUSH & SHAFT MUST BE **0.070 ± 0.020 mm (0.0028 ± 0.0008 inch)**			
C. Idler Gear :- 1011364			
Check gear profile for any scoring , plastic deformation .			
Check bush 1259751 (ID : 81.06 ± 0.01 mm)			
(Note : If the used bush scored & worn , replace with Semi-finished bush then machine as per CAT guidelaine)			
Maintain the correct clearance between the shaft and idler gear and cluster gear. The clearance in diameter between the shaft and idler gear is 0.060 ± 0.020 mm (0.0024 ± 0.0008 inch).			

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D. Idler shaft :- 1011367 Qty. 02			
Check idler shaft ID for any scoring, pitting , rust formation, dent & nicks . Shaft OD : 81.00 ± 0.010 mm)			
Thrust washer (1011368) thickness : 1.90 ± 0.10 mm . No scoring should be there. (SENR 2351)			
Mounting bolt (5/8 " dia) torque : 175 ± 15 lb-ft (SENR5189)			
E . Cam Gear :- 1011369 Qty.02			
Check gear teeth for scoring , damage etc .			
Use new thrust palte (7N3218 Qty.02 ; Bolt 1A2029 Qty.04 & Washer 8M9024 Qty.04 as MANDATORY). Torque for Cam Gear mtg. bolt 270 ± 30 lb ft(SENR5189)			
Torque for thrust plate 3/8 " bolts 40 lb-ft			

Inspected by :	**Checked by :**
Date :	**Date :**

Prepared by : N.Das